

Beyond the Extractive Shortcut: A Cheap, Reference-Free Likelihood–Entailment Metric for Multi-Dimensional Summary Evaluation

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Abstract

Evaluating summaries at scale needs metrics that work without human references and without a call to a large language model for every summary. We propose LNC, a reference-free Likelihood–NLI Composite that scores coherence, consistency, fluency and relevance from two small open models: ten signals from a 1.5B-parameter causal language model and a DeBERTa entailment model, including source-conditioned likelihood, contrasts against minimally perturbed versions of the summary and sentence-level entailment, combined by a per-dimension ridge calibration fitted on 50 SummEval articles. To show that LNC measures more than surface overlap, we evaluate it under a protocol that controls for the extractive shortcut of summarisation benchmarks: on SummEval, a bare bigram copy rate already correlates with human ratings better than ten of the fourteen established metrics we evaluate, so we report every correlation both raw and with the copy rate partialled out, and time every metric on the same machine. On held-out SummEval articles LNC reaches a partial correlation of .350 at 216 ms per summary, above every other reference-free metric, including a local LLM judge (.291) that costs 82 times more, and it lies on the cost–quality Pareto frontier on both axes; only UniEval, which needs a reference for relevance, is ahead (.424). A pre-registered transfer test with the calibration frozen shows that LNC’s consistency judgement generalises: on FRANK CNN/DM it outperforms UniEval and the LLM judge and matches the specialised AlignScore, and on the abstractive FRANK XSum it ties all three. Its relevance judgement does not transfer to RoSE’s content-unit annotations, and controls show that part of the calibration is specific to the systems it was fitted on. Negative results with three alternative designs explain why signals that merely cancel surface overlap lose useful information, and motivate combining likelihood with entailment instead.

Keywords: summarisation evaluation; reference-free metrics; natural language inference; language-model likelihood; extractive shortcut; meta-evaluation; pre-registration

1 Introduction

Automatic summarisation is evaluated far more often than humans can rate it. Every training run, model comparison and ablation needs a score for thousands of candidate summaries, so the field relies on automatic metrics, and those metrics are trusted because they correlate with human judgement on a few meta-evaluation benchmarks. SummEval [8] is the most widely used of them: 100 CNN/DailyMail articles [17], 16 summarisation systems, and expert ratings of coherence, consistency, fluency and relevance for every summary.

The metrics that agree best with these ratings are expensive or need resources that are often missing. Reference-based metrics, from ROUGE [27] to BERTScore [54], need a human-written

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reference summary. Learned evaluators such as UniEval [56] need a reference for some dimensions, and large language model (LLM) judges such as G-Eval [28] need one or more calls to a large model per summary and per dimension. Specialised reference-free metrics such as SummaC [23] and AlignScore [53] are cheap but address factual consistency only. What is missing is a metric that is cheap, needs no reference, and covers all four dimensions.

LNC. We propose LNC, the Likelihood–NLI Composite, which fills this gap with two small open models. A 1.5B-parameter causal language model reads the source and the summary and provides three kinds of evidence: how plausible the summary is given the source, how much more plausible it is than minimally perturbed versions of itself (with swapped sentences, flipped negations, exchanged entities or changed numbers), and how plausible its words are at the points where it departs from the source. An NLI model checks whether every summary sentence is entailed by a nearby passage of the source. A per-dimension ridge regression, fitted on 50 SummEval articles and frozen afterwards, combines these ten signals into four scores. Every signal has prior art; what is new is the selection of signals that remain informative once surface overlap is controlled for, their per-dimension combination, and the evidence on where such a calibrated metric does and does not transfer.

Measuring more than overlap. Evaluating a metric of this kind requires care, because summarisation benchmarks offer an extractive shortcut. The median SummEval summary copies 92% of its token bigrams from the source, and the human ratings follow the amount of copying: summaries that copy more are rated as more consistent, more fluent and, to a lesser degree, more coherent and relevant. Any metric that rewards overlap with the source therefore inherits a correlation it did not earn by assessing quality. The effect is large: the bigram copy rate alone, with no model, no reference and no parameter, correlates with the human ratings better than ten of the fourteen established metrics in our pool. Durmus et al. [6] and Ladhak et al. [24] observed a related effect for factual consistency. We turn it into an evaluation axis for all four dimensions: every correlation is reported both raw and with the copy rate partialled out, so that a metric is credited only for what it measures beyond the shortcut, and every metric is timed on the same machine, so that cost and quality can be compared directly.

Results in brief. On held-out SummEval articles LNC reaches a partial correlation of .350 at 216 ms per summary. It is ahead of every other reference-free metric, including a local LLM judge that costs 82 times more, and it lies on the cost–quality Pareto frontier of both axes; only UniEval, which uses a reference for relevance, is more accurate. A pre-registered test with the calibration frozen shows that LNC’s consistency judgement transfers to FRANK, where it outperforms UniEval and the LLM judge on the CNN/DailyMail part, matches AlignScore, and ties all three on the abstractive XSum part, whereas its relevance judgement does not transfer to RoSE.

Contributions.

- LNC, a reference-free metric for all four dimensions of summary quality, built from a small language model and an NLI model, more accurate than every other reference-free metric we evaluate at a fraction of the cost of an LLM judge.
- An evaluation protocol that credits a metric only for what it measures beyond the extractive shortcut, applied to 23 metrics and controls on SummEval with every metric timed on the same hardware.
- A pre-registered transfer test on FRANK and RoSE with the calibration frozen, showing that the consistency judgement generalises to new systems and annotation schemes while the relevance judgement does not.

- Controls and negative results that bound the claim and explain the design: part of the calibration is specific to the systems it was fitted on, a ridge over existing reference-based metrics reaches the same in-domain score, and three designs that tried to cancel the shortcut inside the language model failed for an identifiable reason.

All code, per-sample scores, the frozen calibration and the scripts that render every table in this paper are publicly available, and the whole study can be regenerated with one command.

2 Related work

Reference-based metrics. The oldest family compares a candidate with one or more human references. Lexical metrics count shared n -grams (BLEU [37], ROUGE [27], METEOR [2]) or characters (chrF [38]); SMART [1] matches sentences instead of tokens. Embedding metrics replace exact matches by similarity in a contextual embedding space: BERTScore [54] aligns tokens greedily and MoverScore [55] solves a transport problem. Trained metrics such as BLEURT [44] and COMET [42] fine-tune an encoder on human judgements. All of them need a reference, which is the resource that is missing whenever summaries are produced at scale.

Reference-free metrics before LLM judges. Several metrics compare the summary with the source instead. SUPERT [12] builds a pseudo-reference from salient source sentences and scores the summary against it with contextual embeddings. BLANC [48] measures how much a summary helps a masked language model reconstruct masked source tokens. SummaQA [43] asks questions generated from the source and checks whether a QA model can answer them from the summary; QAFactEval [9] refines this idea for factual consistency. ESTIME [47] counts summary tokens whose contextual embeddings are mismatched in the source. These methods are cheap and need no reference, but most of them reward overlap with the source by construction.

Likelihood-based scoring. A language model can score a summary by its probability. BARTScore [52] uses a sequence-to-sequence model in several directions, including source to summary; GPTScore [11] prompts a causal model with an instruction. FFLM [20] combines probability changes with and without the source to judge faithfulness. Prism [46] already noted that a paraphraser scores a verbatim copy of its input highly. Perturbation contrasts compare the probability of a text with that of perturbed versions: DeltaScore [51] uses a single perturbation per aspect for story evaluation, and DetectGPT [32] averages over many perturbations to detect machine-generated text. LNC reuses these ideas as features; it does not introduce a new likelihood primitive.

Factual consistency. Inconsistency with the source is the best studied error of abstractive summarisers [31]. FactCC [22] trains a classifier on synthetic errors. SummaC [23] applies an off-the-shelf NLI model to every pair of source and summary sentences and aggregates the entailment scores; its zero-shot variant is one of LNC’s components. AlignScore [53] trains a unified alignment model on many tasks and is a strong published baseline for consistency, which is why we include it. MiniCheck [45] trains small fact-checkers for grounding documents. FRANK [36] provides human factuality annotations with an error typology and is one of our transfer corpora.

Unified evaluators and LLM judges. UniEval [56] casts every dimension as a Boolean question answered by a T5 model trained on pseudo-labelled data. G-Eval [28] asks GPT-4 for a score with chain-of-thought instructions and averages twenty samples; the broader use of LLMs as judges is surveyed by Gu et al. [15]. These evaluators correlate best with human ratings, but the strongest configurations depend on proprietary models or need a reference for some dimensions.

Table 1: Families of summary-evaluation metrics. “Reference” is whether a human reference is needed; cost is per summary on one GPU.

Family	Examples	Reference	Cost	Strength	Weakness
Lexical overlap	BLEU, ROUGE, METEOR, chrF	yes	<5 ms	transparent, free	ignores paraphrase; weak on all dimensions
Embedding similarity	BERTScore, MoverScore, SMART	yes	0.2–1.5 s	robust to paraphrase	needs a reference; weak on consistency
Source-based, pre-LLM	SUPERT, BLANC, SummaQA, ESTIME	no	moderate	no reference	reward overlap with the source
Likelihood	BARTScore, GPSTScore, FFLM	optional	30–300 ms	cheap, graded	reward copying when conditioned on the source
NLI / alignment	SummaC, AlignScore, MiniCheck	no	0.1–0.8 s	targets consistency	little signal for the other dimensions
Unified evaluator	UniEval	for relevance	0.3 s	strong on all dimensions	trained on a related corpus; reference for relevance
LLM judge	G-Eval	no	seconds	strong, flexible	expensive; depends on the judge model
Learned combination	MetaMetrics, LNC	no (LNC)	0.2 s (LNC)	combines complementary signals	needs human ratings to calibrate; may not transfer

Learned metric combinations. MetaMetrics [50] calibrates a combination of existing metrics to human preferences and reports runtime next to correlation. LNC belongs to this family: it is a learned combination calibrated on human ratings. It differs in that its inputs are signals from two small models rather than complete metrics, it is designed and assessed on the shortcut-controlled axis, and its calibration is frozen and tested on corpora it was not fitted on.

Meta-evaluation methodology. Deutsch et al. [5] show that correlations on summarisation benchmarks carry wide confidence intervals and recommend resampling tests, which we use throughout. Gehrmann et al. [13] survey wider problems of evaluation practice. Closest to our motivation, Durmus et al. [6] show that reference-free factuality metrics are confounded with extractiveness and Ladhak et al. [24] describe the trade-off between faithfulness and abstractiveness. The lead bias of news summarisation [21] and the extractive strategies measured by Newsroom [14] explain why the confound is so strong on CNN/DailyMail. We extend the confound analysis from consistency to all four SummEval dimensions and to the whole metric pool. RoSE [30] replaces Likert ratings by atomic content units and is our second transfer corpus.

Table 1 summarises the families, their requirements and their known weaknesses.

3 LNC: a Likelihood–NLI Composite

LNC scores a candidate summary y of a source document x on each of the four SummEval dimensions. It needs no reference. It computes ten features with two small models and maps them to one score per dimension with a linear calibration. Table 2 lists the notation.

Table 2: Notation.

Symbol	Meaning
x, y	source document, candidate summary
y'_k	k -th perturbation of y , $k = 1, \dots, K$
$ y $	number of language-model tokens of y
$P(y x)$	probability of y under the causal LM prompted with x
$P(y)$	probability of y under the same LM without the source
c_j	j -th sentence of y
w	source window: one sentence or two adjacent sentences of x
P_e, P_c	NLI probabilities of entailment and contradiction
$f \in \mathbb{R}^{10}$	feature vector of a candidate
μ, σ	feature means and standard deviations on the development half
w_d	ridge weights for dimension d
$\text{cr}(y, x)$	copy rate: share of the token bigrams of y that occur in x
ρ, ρ_{part}	Spearman correlation, raw and with cr partialled out
$\bar{\rho}, \bar{\rho}_{\text{part}}$	the same, averaged over the four dimensions

3.1 Language-model features

A causal language model (Qwen2.5-1.5B [39]) reads the prompt “**Article:** x **Summary:**” and assigns a log-probability to every token of y . The source prefix is encoded once per article and its key-value cache is reused for every candidate, so the cost of the source is shared by all summaries of an article. From these probabilities LNC takes three features: the length-normalised log-likelihood $\log P(y | x)/|y|$, the total $\log P(y | x)$, and the unconditional $\log P(y)/|y|$, computed with the prompt “**Summary:**” alone. The first is the source-to-summary score of BARTScore [52]; the total also reflects length, which matters for fluency; the unconditional term separates fluency from support by the source.

Perturbation margins. LNC generates $K = 4$ local, meaning-changing edits of y , cycling through four families: swapping a named entity for another entity of the source, changing a number, inserting or removing a negation after an auxiliary verb, and swapping two sentences. Entities are approximated by tokens that appear capitalised inside a source sentence, so no tagger is needed. A family that does not apply to a candidate is skipped, and the random choices are seeded by the sample identifier. From the perturbed log-likelihoods LNC takes three features:

$$m_{\text{order}} = \log P(y | x) - \text{mean}_{k \in \text{order}} \log P(y'_k | x), \quad (1)$$

$$m_{\text{neg}} = \log P(y | x) - \text{mean}_{k \in \text{negation}} \log P(y'_k | x), \quad (2)$$

$$z = \frac{\log P(y | x) - \text{mean}_k \log P(y'_k | x)}{\text{std}_k \log P(y'_k | x) + 10^{-3}}, \quad (3)$$

with a margin set to 0 when its family produced no perturbation, and z set to 0 when fewer than two perturbations exist. The order margin asks whether the sentences of y are in a more plausible order than a swapped version, which is a coherence signal [51]; z normalises the overall contrast by its spread, as DetectGPT does [32].

Splice-point likelihood. SummEval summaries are mostly glued together from copied fragments. LNC marks a word of y as a splice word if it is the first word or if the bigram it closes does not occur in the source; all other words lie inside copied fragments. Word log-probabilities are obtained by summing the log-probabilities of the LM tokens that begin inside the word. Two features are the mean log-probability of splice words and of words inside copied fragments.

Table 3: LNC features and their ridge weights per dimension (standardised features, fitted on the first 50 SummEval articles).

Group	Feature	Coh	Con	Flu	Rel
Likelihood	$\log P(y \mid x)/ y $	.05	.14	.14	.20
	$\log P(y \mid x)$	.19	.02	.36	-.10
	$\log P(y)/ y $	.01	-.03	.04	.10
Perturbation	order margin	.22	.09	.04	.17
	negation margin	-.01	.06	.03	.08
	z-margin	.05	-.03	-.05	.01
NLI	mean g_j	-.29	.13	-.05	-.01
	min g_j	.26	.26	.09	.07
Splice	splice words	.02	.08	.07	.08
	inside copied fragments	.03	.04	-.11	.01

3.2 NLI grounding

An NLI cross-encoder (DeBERTa-v3-large [16] fine-tuned on MNLI, FEVER, ANLI, LingNLI and WANLI [25]) checks every summary sentence c_j against source windows w of one or two adjacent sentences. To bound the cost, each sentence is compared only with the four windows that share the most words with it. The evidence for a sentence is

$$g_j = \max_w (P_e(w, c_j) - P_c(w, c_j)), \quad (4)$$

and LNC takes the mean and the minimum of g_j over sentences. The mean is the zero-shot SummaC construction [23]; the minimum captures the case where a single unsupported sentence makes a summary inconsistent. Unlike likelihood, entailment holds for paraphrase, so this group should not pay a bonus for verbatim copying.

3.3 Calibration

For each dimension d , LNC scores a candidate as

$$\text{LNC}_d(y) = \sum_{i=1}^{10} w_{d,i} \frac{f_i - \mu_i}{\sigma_i}. \quad (5)$$

The means μ and deviations σ are computed on a development set. The weights w_d are fitted by ridge regression [18] with penalty $\alpha = 10$ on the standardised features, with the standardised rank of the human rating as the target. The development set is the first 50 SummEval articles in dataset order (800 summaries); the remaining 50 articles are never used for fitting. The calibration is then frozen and applied unchanged to every other corpus.

Three design decisions were made on the development half only: the ridge was preferred to choosing the best single feature per dimension by article-level cross-validation inside the development half, α was fixed before fitting, and three cost reductions were fixed a priori: the language model runs in bfloat16, $K = 4$ perturbations instead of eight, and four NLI windows per sentence instead of all of them. Table 3 lists the features and the ridge weights.

The weights are readable. Fluency rests on the total log-likelihood, which also favours shorter summaries. Consistency rests on the minimum NLI evidence and the per-token likelihood. Coherence and relevance use the order margin, and coherence contrasts the minimum with the mean NLI evidence, which rewards summaries whose sentences are uniformly supported. Because the features are correlated, single weights should not be read as causal effects; Section 7 measures the contribution of each group by removing it.

Table 4: Human-rated corpora. Copy rate is the median share of summary bigrams that occur in the source.

Corpus	Articles	Systems	Human score	Used as	Copy rate
SummEval	100	16	1–5 Likert, 4 dimensions	dev (1–50) / test (51–100)	.92
FRANK CNN/DM	247	5	share of error-free sentences	consistency	.90
FRANK XSum	249	4	share of error-free sentences	consistency	.21
RoSE CNN/DM	495	11	ACU recall	relevance	.91

4 Experimental setup

4.1 Corpora

Table 4 lists the four human-rated corpora. SummEval [8] is the development and in-domain test corpus. Its 100 CNN/DailyMail articles are split in dataset order: LNC is calibrated on the first 50, and every SummEval number in this paper is computed on the last 50, for all metrics alike. The split is fixed by the order of the public release, and Section 7 repeats the calibration on 200 random article splits.

FRANK [36] and RoSE [30] are the transfer corpora. FRANK annotates factual errors in summaries of CNN/DailyMail and XSum [33] articles; its factuality score is the share of summary sentences without an error, and we treat it as consistency. We report its two parts separately, because XSum summaries are single, highly abstractive sentences. RoSE annotates whether each atomic content unit of a reference is present in a summary; its ACU score measures content coverage, and we treat it as relevance. We use the CNN/DailyMail test set of RoSE. Articles that also occur in SummEval (3 in FRANK, 5 in RoSE) are removed, and so is RoSE’s **gold** system, whose output is the reference that UniEval compares against. Summaries are lowercased and tokenised like SummEval’s; a script rebuilds all four files byte for byte from their public sources.

4.2 Metrics

The pool has 23 metrics and controls (Table 5). Reference-based metrics compare the candidate with each of the 11 SummEval references and keep the maximum. UniEval, G-Eval and LNC produce one score per dimension; every other metric produces a single score that is correlated with all four dimensions.

Several baselines run in configurations smaller than in their original papers, because every metric runs on one machine. G-Eval uses a quantised 7B open model (Qwen2.5-7B-Instruct, 4-bit) with three samples at temperature 0.7 instead of GPT-4 with twenty samples; its results characterise a local LLM judge, not G-Eval with a frontier model. GPTEval uses GPT-2 small (124M parameters) instead of a large GPT-3 model. Our UniEval implementation reads the probability of “Yes” once for the whole summary, without the sentence-level aggregation that the original uses for consistency. NLIG reimplements the zero-shot SummaC construction with a stronger NLI model than the original. AlignScore uses the official AlignScore-large weights and inference procedure, with our own sentence splitter.

LGS is the lead-biased embedding grounding score of our earlier, unpublished study, kept as a data point for its signal class. CCM, PSC and SPL are three reference-free metrics we built while searching for a signal that is not a copy detector; they failed as metrics and are analysed in Section 7. Their signals, together with NLIG, are the components of LNC.

Table 5: Metric pool. “Ref” marks metrics that need a human reference. Parameters count the neural models used; “—” marks metrics without one.

Metric	Model	Params	Ref	Score
BLEU, ROUGE-L, chrF, METEOR	—	—	yes	n -gram or character overlap with the reference
SMART-String	—	—	yes	sentence-matched ROUGE
SMART-Model	nomic-embed-text [34]	137M	yes	sentence-matched embedding cosine
EmbedScorer	nomic-embed-text	137M	yes	cosine of whole-text embeddings
BERTScore	RoBERTa-large [29]	355M	yes	token-alignment F1
MoverScore	RoBERTa-large	355M	yes	transport distance of token embeddings
BARTScore	BART-large-CNN [26]	406M	yes	$\log P(\text{ref} y)$ per token
GPTScore	GPT-2 small [40]	124M	yes	$\log P(y \text{ref})$ per token
UniEval	T5-large [41]	780M	relevance	$P(\text{Yes})$ to a dimension question
G-Eval	Qwen2.5-7B-Instruct, 4-bit	7.6B	no	judge score, 3 samples
AlignScore	RoBERTa-large	355M	no	alignment probability, chunked source
NLIG (SummaC-ZS)	DeBERTa-v3-large	435M	no	mean NLI evidence over all windows
LGS	nomic-embed-text	137M	no	lead-weighted sentence grounding, $\lambda = 0.5$
Lead-3, Lead-5	—	—	no	ROUGE-L against the first k source sentences
Lead-3 (whole)	—	—	no	ROUGE-L against the first 3 sentences as one block
CCM	Qwen2.5-1.5B	1.5B	no	perturbation margin, $K = 8$
PSC	Qwen2.5-1.5B + 7B paraphraser	1.5B + 7.6B	no	CCM margin conditioned on a paraphrased source
SPL	Qwen2.5-1.5B	1.5B	no	likelihood of splice words
LNC	Qwen2.5-1.5B + DeBERTa-v3-large	1.9B	no	calibrated composite of ten features

4.3 Evaluation protocol

Correlation. The summary-level correlation of a metric with a human dimension is the Spearman correlation between metric scores and human ratings over all summaries of the evaluation set. Pooling summaries across articles is common practice; per-article correlations are used in the Friedman test below. $\bar{\rho}$ averages the four SummEval dimensions.

The shortcut-controlled axis. The copy rate $\text{cr}(y, x)$ is the share of the candidate’s token bigrams that occur in the source after lowercasing and splitting off punctuation. The partial correlation removes the rank correlation that both the metric m and the human rating h share with the copy rate:

$$\rho_{\text{part}}(m, h) = \frac{\rho(m, h) - \rho(m, \text{cr}) \rho(h, \text{cr})}{\sqrt{(1 - \rho(m, \text{cr})^2)(1 - \rho(h, \text{cr})^2)}}. \quad (6)$$

ρ_{copy} is the correlation of the metric with the copy rate. We use bigrams because unigram overlap is saturated by function words. The partial axis does not claim that copying is bad; it asks which part of a metric’s agreement with humans is not explained by copying, the cheapest signal there is.

Significance. Confidence intervals and paired comparisons use the cluster bootstrap over articles [5, 7], because the summaries of one article share a source. A paired comparison resamples articles 5,000 times, recomputes both correlations on each resample, and reports the mean difference, its 95% interval and a two-sided p -value; on the partial axis the copy rate is partialled out on every resample. p -values are adjusted with the Benjamini–Hochberg procedure [3] over all cells of a table, and a difference is called significant when $q < 0.05$ and the interval excludes zero. Across the whole pool we run a Friedman test [4, 10] with articles as blocks and per-article Spearman correlations as responses, followed by Wilcoxon signed-rank tests [49] of LNC against every other metric with Holm correction [19].

Cost. All metrics run on one ASUS Ascent GX10 workstation with an NVIDIA GB10 (Blackwell) and 128 GB of unified memory: the neural metrics on its GPU through a single Python model server, the lexical metrics on its CPU, and G-Eval and the embedding metrics through Ollama [35]. Every model is loaded before timing starts. Cost is wall-clock time per summary over the whole corpus, including all model calls; for per-dimension metrics it is the sum over dimensions. Wall-clock time depends on the hardware, but the comparisons we draw involve differences of one to four orders of magnitude.

Pre-registration. The transfer protocol of Section 6, including the corpora, the baselines, the primary axis and the success criterion, was written into the project documentation on 26 September 2026, before any metric was run on FRANK or RoSE. It was committed to the public repository together with the results, so the order is documented by our records rather than by an external registry; later edits of that text only renamed files and commands, and AlignScore, added afterwards, is reported outside the criterion.

Table 6: Correlation with human ratings beyond the extractive shortcut on SummEval, held-out last 50 articles. The copy rate of a candidate is the fraction of its token 2-grams that occur in the source; it requires no model, no reference and no hyperparameter. $\bar{\rho}$ is the summary-level Spearman correlation with human ratings averaged across the four dimensions (the last four columns give the partial value per dimension), $\bar{\rho}_{\text{part}}$ the same quantity with the copy rate partialled out of both sides, and ρ_{copy} the correlation between the metric and the copy rate (mean over its per-dimension scorers). Brackets give the 95% cluster-bootstrap interval over articles. The final row is the copy rate scored as if it were a metric. Pareto status is given on both axes.

Metric	ms/sample	$\bar{\rho}$	$\bar{\rho}_{\text{part}}$	ρ_{copy}	Coh	Con	Flu	Rel	Pareto
UniEval	343.00	.473	.424 [.386, .459]	.300	.550	.289	.426	.429	raw, partial
LNC	216.02	.433	.350 [.313, .385]	.550	.350	.370	.340	.339	raw, partial
G-Eval	17823.14	.327	.291 [.251, .333]	.229	.315	.408	.149	.292	—
BERTScore	213.89	.269	.269 [.214, .319]	.027	.425	.105	.128	.420	partial
NLIG	477.80	.274	.224 [.174, .275]	.255	.134	.413	.224	.124	—
AlignScore	128.09	.282	.200 [.156, .246]	.399	.118	.318	.295	.069	partial
GPTScore	32.83	.290	.194 [.140, .246]	.448	.360	.063	.060	.294	partial
EmbedScorer	535.47	.222	.175 [.118, .228]	.207	.184	.086	.104	.325	—
LGS	93.85	.299	.167 [.112, .224]	.654	.210	.146	.091	.220	raw
PSC	546.93	.236	.159 [.103, .215]	.339	.211	.144	.121	.160	—
METEOR	1.78	.125	.156 [.097, .215]	-.091	.202	.125	.113	.181	partial
Lead-5 (mean-of-max)	0.08	.292	.147 [.091, .209]	.708	.190	.135	.062	.202	raw, partial
CCM	190.86	.248	.142 [.090, .198]	.473	.171	.167	.117	.115	—
Lead-3 (whole block)	0.02	.261	.139 [.083, .197]	.549	.165	.097	.080	.215	raw, partial
Lead-3 (mean-of-max)	0.05	.270	.137 [.081, .195]	.611	.174	.118	.080	.173	raw
SMART-Model	1498.21	.173	.136 [.077, .199]	.158	.126	.097	.093	.228	—
MoverScore	406.69	.089	.129 [.066, .183]	-.129	.223	.036	.055	.201	—
ROUGE-L	0.87	.140	.127 [.060, .190]	.066	.128	.113	.099	.167	—
SPL	25.62	.092	.120 [.071, .168]	-.088	.145	.067	.176	.093	—
ChrF	4.65	.164	.117 [.056, .175]	.195	.091	.064	.072	.241	—
BARTScore	303.79	.116	.100 [.034, .166]	.071	.080	.019	.111	.188	—
BLEU	1.50	.117	.087 [.026, .141]	.126	.053	.072	.019	.203	—
SMART-String	1.38	.056	.074 [.014, .126]	-.051	-.024	.097	.120	.102	—
Copy rate (no metric)	0.00	.271	.000	1.000	.000	.000	.000	.000	<i>diagnostic</i>

Table 7: Paired bootstrap comparison of LNC against baselines on SummEval, held-out last 50 articles, copy-partialled summary-level Spearman correlations. Each cell reports $\Delta\rho$ (ours minus baseline) with 95% CI in brackets, the raw two-sided p -value, and q , the Benjamini-Hochberg adjusted p -value over all 36 (baseline, dimension) cells of this table. Bold marks improvements that survive the correction ($q < 0.05$ and CI excludes 0).

Baseline	Coh	Con	Flu	Rel
UniEval	-.200 [-.285, -.113] $p < .001$ $q < .001$	+.082 [+.034, +.129] $p < .001$ $q < .001$	-.086 [-.144, -.027] $p=0.004$ $q=0.007$	-.089 [-.188, +.011] $p=0.088$ $q=0.122$
G-Eval	+.034 [-.061, +.127] $p=0.484$ $q=0.498$	-.039 [-.102, +.028] $p=0.244$ $q=0.274$	+.190 [+.116, +.269] $p < .001$ $q < .001$	+.046 [-.058, +.153] $p=0.384$ $q=0.406$
AlignScore	+.231 [+.136, +.325] $p < .001$ $q < .001$	+.051 [-.022, +.125] $p=0.173$ $q=0.207$	+.044 [-.027, +.123] $p=0.236$ $q=0.274$	+.269 [+.167, +.378] $p < .001$ $q < .001$
NLIG	+.217 [+.125, +.311] $p < .001$ $q < .001$	-.042 [-.091, +.008] $p=0.099$ $q=0.132$	+.114 [+.045, +.190] $p < .001$ $q=0.002$	+.216 [+.112, +.317] $p < .001$ $q < .001$
LGS	+.141 [+.040, +.245] $p=0.007$ $q=0.012$	+.224 [+.163, +.287] $p < .001$ $q < .001$	+.248 [+.145, +.350] $p < .001$ $q < .001$	+.121 [+.017, +.231] $p=0.024$ $q=0.035$
BERTScore	-.073 [-.165, +.024] $p=0.137$ $q=0.176$	+.264 [+.170, +.363] $p < .001$ $q < .001$	+.209 [+.117, +.306] $p < .001$ $q < .001$	-.077 [-.185, +.030] $p=0.165$ $q=0.205$
GPTScore	-.009 [-.114, +.103] $p=0.859$ $q=0.859$	+.305 [+.226, +.388] $p < .001$ $q < .001$	+.278 [+.185, +.372] $p < .001$ $q < .001$	+.049 [-.059, +.152] $p=0.362$ $q=0.395$
BARTScore	+.269 [+.171, +.368] $p < .001$ $q < .001$	+.347 [+.254, +.440] $p < .001$ $q < .001$	+.229 [+.137, +.323] $p < .001$ $q < .001$	+.152 [+.008, +.295] $p=0.038$ $q=0.055$
Lead-5	+.160 [+.055, +.268] $p=0.003$ $q=0.006$	+.235 [+.171, +.300] $p < .001$ $q < .001$	+.276 [+.190, +.360] $p < .001$ $q < .001$	+.138 [+.020, +.253] $p=0.020$ $q=0.031$

Table 8: Friedman test on SummEval: blocks are the 50 articles, treatments the 19 metrics, and the response is each metric’s within-article Spearman correlation with the human ratings. Cells give the average rank (higher is better). An asterisk marks a metric whose Wilcoxon signed-rank comparison with LNC is significant after Holm correction ($p_{\text{Holm}} < 0.05$). The last rows give the Friedman statistic and the Nemenyi critical difference (CD, $\alpha = 0.05$).

Metric	Coh	Con	Flu	Rel
UniEval	16.54*	11.78*	14.91	14.63
LNC	13.88	15.67	14.75	12.49
G-Eval	13.58	15.82	6.73*	13.76
AlignScore	10.40*	14.77*	14.19	9.21*
LGS	10.57*	12.17*	12.04*	10.12
NLIG	8.72*	13.97*	12.09*	8.96*
GPTScore	13.13	8.97*	10.52*	9.37*
Lead-5 (sent)	9.43*	11.78*	11.51*	8.49*
Lead-3 (sent)	9.73*	10.94*	11.20*	8.74*
BERTScore	13.06	6.84*	8.99*	11.56
EmbedScorer	8.96*	8.66*	9.62*	12.45
SMART-Model	8.78*	9.26*	9.39*	10.30
BARTScore	8.72*	7.28*	8.23*	11.09
ChrF	8.00*	7.70*	8.20*	10.71
ROUGE-L	8.51*	8.54*	9.19*	8.23*
METEOR	8.15*	6.77*	7.38*	7.92*
BLEU	6.25*	7.28*	7.32*	7.63*
SMART-String	5.56*	7.09*	7.62*	7.33*
MoverScore	8.03*	4.71*	6.12*	7.01*
χ_F^2	227.1	314.1	206.5	137.6
CD	3.96	3.96	3.96	3.96

5 Results on SummEval

All numbers in this section are computed on the 50 held-out SummEval articles (800 summaries). Table 6 is the main result: for every metric it gives the cost, the raw and the partialled mean correlation, the correlation with the copy rate, the partialled correlation per dimension, and whether the metric is on the cost-quality Pareto frontier of either axis. Appendix A gives the raw per-dimension correlations with bootstrap intervals and the system-level correlations.

5.1 The extractive shortcut

The last row of Table 6 scores the copy rate as if it were a metric. Its mean raw correlation of .271 is higher than that of BERTScore (.269), EmbedScorer, SMART-Model, MoverScore, BARTScore and all lexical metrics: ten of the fourteen baselines are beaten by a ratio of shared bigrams. The human ratings are themselves correlated with the copy rate (Spearman .21 for coherence, .38 for consistency, .31 for fluency and .20 for relevance over all of SummEval), and every metric that looks at the source inherits part of that correlation. The lead-window controls, which match the summary against the first three or five source sentences with ROUGE-L, are the clearest case. Lead-5 reaches a raw $\bar{\rho}$ of .292, higher than BERTScore, GPTScore and AlignScore, but its correlation with the copy rate is .708 and its partialled $\bar{\rho}$ falls to .147.

Controlling for the shortcut reorders the pool. Metrics that see only the reference, such as BERTScore ($\rho_{\text{copy}} = .027$), lose nothing, because copying from the source cannot leak into them. Source-based similarity metrics lose most: LGS drops from .299 to .167, CCM from .248 to .142, and the lead controls by half. The learned and model-based evaluators lose least: UniEval drops from .473 to .424 and G-Eval from .327 to .291. The partial axis also separates metrics that the raw axis puts close together: LGS and Lead-5 have raw correlations close to those of GPTScore and AlignScore (.299 and .292 against .290 and .282), but lower partial ones (.167 and .147 against .194 and .200).

5.2 LNC against the pool

LNC reaches a raw $\bar{\rho}$ of .433 and a partial $\bar{\rho}$ of .350 (95% interval [.313, .385]). It is second to UniEval on both axes and ahead of every other metric, including the local G-Eval judge (.327 raw, .291 partial). Its loss from raw to partial (.083) is larger than UniEval’s (.049) and G-Eval’s (.036), and its correlation with the copy rate (.550) is high: LNC still uses copying, but most of its agreement with the human ratings survives when copying is controlled for.

Per dimension, LNC is balanced where its competitors are not. Its partial correlations are .350, .370, .340 and .339 for coherence, consistency, fluency and relevance. UniEval is much stronger on coherence (.550) and weaker on consistency (.289). G-Eval is strong on consistency (.408) and weak on fluency (.149). BERTScore is strong on coherence and relevance (.425, .420) and close to zero on consistency and fluency. AlignScore and NLIG are consistency metrics: .318 and .413 on consistency, at most .295 elsewhere.

Table 7 tests these differences on the partial axis with paired bootstrap comparisons over the 36 cells of LNC against nine baselines. LNC is significantly better in 22 cells, tied in 12 and worse in 2, both against UniEval: on coherence (−.200) and fluency (−.086). Against UniEval it is better on consistency (+.082), and the difference on relevance is not significant. Against G-Eval it is better on fluency (+.190) and tied on the other three dimensions. Against AlignScore and NLIG it ties on consistency, their own dimension, and is better on coherence and relevance. It is better than LGS, Lead-5 and BARTScore on all four dimensions. On the raw axis the picture is the same, with 25 wins, 10 ties and one loss, to UniEval on coherence (the table is rendered by the released code).

Table 9: Correlation with human ratings beyond the extractive shortcut on FRANK (CNN/DM). The copy rate of a candidate is the fraction of its token 2-grams that occur in the source; it requires no model, no reference and no hyperparameter. $\bar{\rho}$ is the summary-level Spearman correlation with human ratings on the consistency dimension, $\bar{\rho}_{\text{part}}$ the same quantity with the copy rate partialled out of both sides, and ρ_{copy} the correlation between the metric and the copy rate (mean over its per-dimension scorers). Brackets give the 95% cluster-bootstrap interval over articles. The final row is the copy rate scored as if it were a metric. Pareto status is given on both axes.

Metric	ms/sample	$\bar{\rho}$	$\bar{\rho}_{\text{part}}$	ρ_{copy}	Pareto
LNC	220.07	.622	.516 [.472, .559]	.575	raw, partial
AlignScore	195.91	.602	.501 [.452, .547]	.501	raw, partial
NLIG	777.50	.578	.488 [.441, .535]	.425	—
G-Eval	2562.69	.539	.446 [.395, .496]	.407	—
UniEval	200.40	.545	.445 [.398, .491]	.438	—
LGS	178.61	.381	.180 [.127, .234]	.606	raw, partial
Lead-5 (mean-of-max)	0.09	.325	.121 [.066, .178]	.566	raw, partial
Lead-3 (mean-of-max)	0.05	.255	.093 [.035, .148]	.434	raw, partial
Copy rate (no metric)	0.00	.413	.000	1.000	<i>diagnostic</i>

5.3 Rank tests over the whole pool

The Friedman test (Table 8) treats each article as a block and ranks the metrics by their within-article correlation with the human ratings. The null hypothesis of equal average ranks is rejected on every dimension (χ^2_F between 137.6 and 314.1 with 18 degrees of freedom, $p < .001$). LNC and G-Eval have the two highest average ranks on consistency, and LNC has the second highest on coherence and fluency, after UniEval. Holm-corrected Wilcoxon tests separate LNC from UniEval on coherence and consistency, where UniEval ranks higher on coherence and lower on consistency, and from G-Eval on fluency, but not from G-Eval on coherence, consistency or relevance, nor from UniEval on fluency and relevance. Because this test uses within-article correlations, it is insensitive to differences between articles; that it agrees with the pooled analysis shows that LNC’s advantage is not an artefact of pooling.

5.4 Cost

LNC needs 216 ms per summary: 106 ms for the language model (one source pass per article and five candidate variants), 85 ms for NLI and 25 ms for the per-token pass of the splice features. That is less than UniEval with all four dimensions (343 ms), about the same as BERTScore (214 ms), and 82 times less than the local G-Eval judge (17.8 s for four dimensions with three samples each). LNC is on the Pareto frontier of both axes in Table 6: no cheaper metric correlates better, raw or partialled. On the partial axis the frontier runs from the lead controls and METEOR through GPTScore, AlignScore and BERTScore to LNC and UniEval; on the raw axis it consists of the lead controls, LGS, LNC and UniEval. The comparison with UniEval needs one qualification: UniEval scores relevance against the reference, which LNC does not have. Among reference-free metrics LNC is the most accurate at any cost below that of an LLM judge, and more accurate than the LLM judge we could run.

Table 10: Correlation with human ratings beyond the extractive shortcut on FRANK (XSum). The copy rate of a candidate is the fraction of its token 2-grams that occur in the source; it requires no model, no reference and no hyperparameter. $\bar{\rho}$ is the summary-level Spearman correlation with human ratings on the consistency dimension, $\bar{\rho}_{\text{part}}$ the same quantity with the copy rate partialled out of both sides, and ρ_{copy} the correlation between the metric and the copy rate (mean over its per-dimension scorers). Brackets give the 95% cluster-bootstrap interval over articles. The final row is the copy rate scored as if it were a metric. Pareto status is given on both axes.

Metric	ms/sample	$\bar{\rho}$	$\bar{\rho}_{\text{part}}$	ρ_{copy}	Pareto
AlignScore	41.08	.296	.293 [.238, .343]	.067	raw, partial
LNC	117.26	.276	.280 [.217, .338]	-.041	—
G-Eval	1948.99	.264	.264 [.205, .318]	.006	—
UniEval	118.33	.261	.264 [.209, .314]	-.028	—
NLIG	183.32	.260	.255 [.190, .316]	.106	—
LGS	115.66	.116	.106 [.044, .164]	.203	—
Lead-5 (mean-of-max)	0.03	.100	.081 [.010, .147]	.415	raw, partial
Lead-3 (mean-of-max)	0.02	.073	.054 [-.015, .120]	.367	raw, partial
Copy rate (no metric)	0.00	.063	.000	1.000	<i>diagnostic</i>

Table 11: Correlation with human ratings beyond the extractive shortcut on RoSE (CNN/DM). The copy rate of a candidate is the fraction of its token 2-grams that occur in the source; it requires no model, no reference and no hyperparameter. $\bar{\rho}$ is the summary-level Spearman correlation with human ratings on the relevance dimension, $\bar{\rho}_{\text{part}}$ the same quantity with the copy rate partialled out of both sides, and ρ_{copy} the correlation between the metric and the copy rate (mean over its per-dimension scorers). Brackets give the 95% cluster-bootstrap interval over articles. The final row is the copy rate scored as if it were a metric. Pareto status is given on both axes.

Metric	ms/sample	$\bar{\rho}$	$\bar{\rho}_{\text{part}}$	ρ_{copy}	Pareto
UniEval	47.40	.217	.245 [.206, .284]	-.160	raw, partial
G-Eval	2070.24	.230	.238 [.192, .281]	-.040	raw
Lead-3 (mean-of-max)	0.05	.201	.151 [.096, .206]	.532	raw, partial
LNC	215.04	.182	.148 [.106, .191]	.301	—
Lead-5 (mean-of-max)	0.07	.195	.138 [.086, .189]	.655	—
AlignScore	279.14	.055	.025 [-.020, .070]	.222	—
LGS	153.72	.124	.022 [-.023, .070]	.786	—
NLIG	1068.59	.057	.017 [-.023, .056]	.285	—
Copy rate (no metric)	0.00	.141	.000	1.000	<i>diagnostic</i>

Table 12: Paired bootstrap comparison of LNC against baselines on FRANK (CNN/DM), copy-partialled summary-level Spearman correlations. Each cell reports $\Delta\rho$ (ours minus baseline) with 95% CI in brackets, the raw two-sided p -value, and q , the Benjamini-Hochberg adjusted p -value over all 7 (baseline, dimension) cells of this table. Bold marks improvements that survive the correction ($q < 0.05$ and CI excludes 0).

Baseline	Con
AlignScore	+.015 [-.031, +.063] $p=0.545$ $q=0.545$
NLIG	+.028 [-.005, +.062] $p=0.106$ $q=0.124$
LGS	+.336 [+.273, +.400] $p < .001$ $q < .001$
Lead-3	+.423 [+.353, +.495] $p < .001$ $q < .001$
Lead-5	+.395 [+.325, +.465] $p < .001$ $q < .001$
UniEval	+.071 [+.030, +.111] $p < .001$ $q < .001$
G-Eval	+.070 [+.023, +.119] $p=0.005$ $q=0.007$

Table 13: Paired bootstrap comparison of LNC against baselines on RoSE (CNN/DM), copy-partialled summary-level Spearman correlations. Each cell reports $\Delta\rho$ (ours minus baseline) with 95% CI in brackets, the raw two-sided p -value, and q , the Benjamini-Hochberg adjusted p -value over all 7 (baseline, dimension) cells of this table. Bold marks improvements that survive the correction ($q < 0.05$ and CI excludes 0).

Baseline	Rel
AlignScore	+.124 [+.062, +.185] $p < .001$ $q < .001$
NLIG	+.132 [+.076, +.188] $p < .001$ $q < .001$
LGS	+.127 [+.063, +.192] $p < .001$ $q < .001$
Lead-3	-.002 [-.068, +.065] $p=0.956$ $q=0.956$
Lead-5	+.012 [-.054, +.077] $p=0.709$ $q=0.827$
UniEval	-.096 [-.147, -.046] $p < .001$ $q=0.001$
G-Eval	-.089 [-.145, -.033] $p < .001$ $q=0.001$

6 Pre-registered transfer to FRANK and RoSE

A metric calibrated on one benchmark may only have learned that benchmark: its systems, its annotators and its rating scale. We therefore tested whether the SummEval calibration of LNC transfers to two corpora with different systems and annotation schemes. The protocol below was fixed before any metric was run on them.

6.1 Protocol

LNC is applied with the frozen SummEval calibration, without refitting and without changing any feature. On FRANK its consistency scorer is correlated with the factuality score, and on RoSE its relevance scorer with the ACU score. The primary axis is the summary-level Spearman correlation with the copy rate partialled out; the raw correlation is secondary. The baselines are the cheap reference-free metrics NLIG, LGS, Lead-3 and Lead-5, together with UniEval and G-Eval for the matching dimension. G-Eval runs once with greedy decoding to fit the compute budget. The success criterion was stated as follows: LNC transfers if, on both FRANK CNN/DM and RoSE, (i) its partial correlation is significantly higher than that of the best cheap reference-free baseline in a paired bootstrap over articles, with Benjamini–Hochberg correction within the table of each corpus, and (ii) it is not significantly worse than G-Eval. FRANK XSum was declared an out-of-domain stress test outside the criterion, because its single-sentence summaries leave the order margin, LNC’s most important perturbation feature, at zero.

AlignScore was added after the pre-registered run, as the strongest published consistency baseline we could run. It is reported in the tables but plays no role in the criterion. With it in the family of tests, the BH-adjusted q -values change slightly; none of the conclusions below depends on it.

6.2 Results

Tables 9–11 give the raw and partial correlations on the three transfer sets, and Tables 12 and 13 the paired comparisons on the partial axis.

FRANK CNN/DM. LNC reaches a partial correlation of .516 with the factuality score (raw .622), the highest of all metrics. It is significantly better than UniEval (+.071) and G-Eval (+.070), and far better than LGS and the lead controls. Its advantage over NLIG, the best cheap reference-free baseline, is +.028 with a 95% interval of $[-.005, +.062]$ and is not significant ($q = .106$ in the pre-registered family, $q = .124$ with AlignScore added). It is also tied with AlignScore (+.015). Condition (ii) holds, but condition (i) fails.

FRANK XSum. On the abstractive XSum summaries the copy rate carries almost no signal (raw correlation with factuality .063), so raw and partial correlations coincide. LNC reaches .280 and ties NLIG (.255), UniEval (.264), G-Eval (.264) and AlignScore (.293). It remains far ahead of LGS and the lead controls, which depend on copying.

RoSE. On content coverage LNC transfers poorly. Its partial correlation of .148 ties Lead-3 (.151) and Lead-5 (.138) and is significantly lower than that of UniEval (−.096) and G-Eval (−.089). It is still significantly better than NLIG, AlignScore and LGS, whose correlation with the ACU score is close to zero once copying is controlled for. Both conditions fail.

Verdict. By the pre-registered criterion, LNC does not transfer. The result is nevertheless informative, because the two calibrated scorers behave differently. The consistency scorer, fitted on SummEval’s Likert ratings, ranks FRANK summaries better than UniEval and the local G-Eval judge and as well as AlignScore, a model trained specifically for factual alignment; most

Table 14: Held-out copy-partialled $\bar{\rho}$ of the LNC ridge recipe on SummEval over 200 random splits (mean and 2.5–97.5 percentiles). Controls apply the same ridge to existing signals; the last block refits LNC without one feature group. The system split fits on 8 of the 16 systems and 50 articles and tests on the other 50 articles; unseen systems score lower in 135 of 200 splits, while zero-shot UniEval does not change.

Configuration	$\bar{\rho}_{\text{part}}$
LNC, random article splits	.322 [.295, .344]
Ridge on cheap reference-free signals	.222 [.186, .251]
Ridge on existing metrics incl. reference-based	.322 [.290, .352]
Ridge on UniEval’s four scorers	.421 [.390, .452]
LNC, fit on 8 systems: same systems	.316 [.195, .412]
LNC, fit on 8 systems: unseen systems	.269 [.153, .350]
UniEval (zero-shot): same systems	.408 [.325, .480]
UniEval (zero-shot): unseen systems	.411 [.329, .490]
LNC without likelihood features	.305 [.276, .334]
LNC without perturbation margins	.280 [.248, .308]
LNC without NLI grounding	.299 [.271, .328]
LNC without splice-point likelihood	.319 [.291, .343]

of this comes from its NLI features, since NLIG alone reaches .488. The relevance scorer does not carry over from SummEval’s relevance ratings to RoSE’s atomic content units: on RoSE, a lead-sentence heuristic does as well. Relevance in SummEval is judged against the source by annotators who read it, whereas the ACU score measures coverage of a reference; a reference-free metric calibrated on the former has no access to the latter, while UniEval reads the reference for relevance and G-Eval reads the full source with a much larger model. FRANK and RoSE have now served as test sets; any change to LNC prompted by these results would have to be validated on a further corpus.

7 Analysis

7.1 How robust is the in-domain result?

Table 14 refits the LNC calibration on 200 random 50/50 splits of the SummEval articles and evaluates each fit on the held-out half.

The canonical split is favourable. Over random splits LNC reaches a partial $\bar{\rho}$ of .322 (2.5–97.5 percentiles .295–.344), against .350 on the split in dataset order. The dataset-order split is therefore on the favourable side of the distribution, and .322 is the better estimate of LNC’s in-domain accuracy. It remains above every reference-free metric in Table 6 and below UniEval.

The same ridge over existing metrics. A calibrated combination needs a calibrated control. Fitting the same ridge to cheap reference-free signals (copy rate, length, Lead-5, LGS and NLIG) gives .222, so LNC’s own signals add about .10. Fitting it to existing metrics that include reference-based ones (BERTScore, GPTScore and BARTScore, besides copy rate, length, LGS and NLIG) gives .322, the same as LNC, and fitting it to UniEval’s four scorers gives .421. A learned combination of metrics that read the reference therefore matches LNC in domain; what LNC contributes is the same level without a reference, from one pair of small models.

Unseen systems. The 16 SummEval systems occur in both halves of every article split, so the evaluation above holds out articles but not systems. To test generalisation to new systems, we

Table 15: Source-conditioned likelihood signals on all 100 SummEval articles: mean raw and partial Spearman correlation over the four dimensions, and correlation with the copy rate. $\tilde{x}$ is a paraphrase of the source written by an instruction-tuned LLM.

Metric	Signal	$\bar{\rho}$	$\bar{\rho}_{\text{part}}$	ρ_{copy}
Likelihood	$\log P(y x) / y $	.353	.233	.741
CCM	$\log P(y x) - \text{mean}_k \log P(y'_k x)$	.224	.123	.429
CCM, normalised	$z\text{-margin}$	.221	.172	.208
PSC likelihood	$\log P(y \tilde{x}) / y $	.260	.223	.179
PSC	$\log P(y \tilde{x}) - \text{mean}_k \log P(y'_k \tilde{x})$	.225	.152	.314
Copy advantage	$(\log P(y x) - \log P(y \tilde{x})) / y $	.151	-.007	.571
SPL	mean $\log P$ of splice words	.123	.139	-.038
SPL, control	mean $\log P$ inside copied fragments	.302	.193	.527
NLIG	mean NLI evidence	.256	.189	.315

fitted LNC on 8 randomly chosen systems in 50 articles and evaluated it on the other 50 articles, once on the same 8 systems and once on the other 8. On the same systems LNC reaches .316, on unseen systems .269, and it is lower on unseen systems in 135 of 200 splits. UniEval, which is not fitted on SummEval, reaches .408 and .411 on the same subsets, so the drop is not caused by the unseen systems being harder to rank. Part of LNC’s calibration is specific to the systems it was fitted on. The transfer results of Section 6 show the same limitation from a different direction.

Feature groups. Removing one group of features and refitting shows what each group contributes. Without the perturbation margins the partial $\bar{\rho}$ falls from .322 to .280, without NLI to .299, without the likelihood features to .305, and without the splice features to .319. The perturbation margins are the most valuable group although, as the next subsection shows, they failed as a metric on their own. The splice features add almost nothing.

7.2 Why removing the shortcut from likelihood fails

LNC was not the first design we tried. Three earlier metrics aimed to remove the extractive shortcut from a source-conditioned language model directly, by cancelling the likelihood bonus that copied text receives. All three failed, and they failed in an instructive way. Table 15 summarises them over all 100 SummEval articles, since none of these signals is fitted on human ratings.

Perturbation margins (CCM). The idea was that a perturbation of a summary copies the source as much as the summary itself, so the copy bonus should cancel in the difference of their log-likelihoods. It does not. The margin keeps a correlation of .429 with the copy rate and its partial $\bar{\rho}$ is only .123, below the plain likelihood. With the source in context, the language model predicts copied spans almost deterministically, so perturbing a copied span costs much more likelihood than perturbing a paraphrased one, and the margin rewards copying. The families differ: entity swaps and number changes carry almost no partial signal for consistency (.096 and .070), which they were meant to detect, while sentence swaps carry a clear signal for coherence (.281) and relevance (.265). This is why the order margin, not the overall margin, became an LNC feature.

Paraphrased-source conditioning (PSC). If the copy bonus comes from verbatim overlap with the conditioning text, conditioning on a paraphrase $\tilde{x}$ of the source should remove it. We had an instruction-tuned LLM rewrite each source in chunks of three sentences while keeping every fact; the paraphrase shares 21% of its bigrams with the source. The mechanism is confirmed: the

correlation of the likelihood with the copy rate falls from .741 to .179, and the copy advantage, the per-token difference between conditioning on x and on $\tilde{x}$, correlates .571 with the copy rate and has no partial correlation with the human ratings at all ($-.007$). The likelihood with a paraphrased source is therefore much less of a copy detector. But it is not a better metric: its partial $\bar{\rho}$ falls slightly, from .233 to .223, because the paraphrase also removes useful signal and occasionally adds content of its own. Removing copying from a signal removes copying, not the confound.

Splice-point likelihood (SPL). The median SummEval summary copies 92% of its bigrams, so we hypothesised that quality is decided where a summary departs from verbatim copying: at the splice points between copied fragments. Scoring only those words removes the copy correlation entirely ($-.038$), but the partial correlation is low (.139). The words inside copied fragments, the control, carry more signal (.193). We verified that the per-word log-probabilities sum back to the sequence log-likelihood, so this is not an implementation artefact.

Together these results explain LNC’s design. Every attempt to make a likelihood signal blind to copying also made it blind to part of what humans reward. LNC instead keeps the copy-sensitive likelihood features, adds signals of a different kind, the order margin and NLI entailment, and lets the calibration weigh them per dimension. Its held-out partial correlation (.350) is higher than that of any single signal in Table 15, while its correlation with the copy rate (.550) remains high.

7.3 Sensitivity of embedding metrics to the platform

The scores of all metrics in this paper were computed on one machine. An earlier run of the baselines on an Apple-silicon machine, preserved in the history of our repository, allows a check of how portable the scores are. The lexical metrics, BERTScore, MoverScore, BARTScore, GPTScore and UniEval agreed to within $5 \cdot 10^{-6}$ between the two platforms, and G-Eval, which samples at temperature 0.7, reached nearly the same correlations with different samples. The three metrics that use `nomic-embed-text` through Ollama did not: the Spearman correlation between their scores on the two platforms was .69 for LGS, .45 for SMART-Model and .41 for EmbedScorer, and the full-set partial $\bar{\rho}$ of LGS moved from .198 to .157. On each platform the embeddings were deterministic and semantically sensible, so the difference comes from the runtime and its version rather than from a bug. Embedding metrics served through a local LLM runtime should be reported with the runtime version and hardware, and comparisons between them across platforms should be avoided.

8 Discussion

What LNC is. LNC is a reference-free metric that scores all four SummEval dimensions from two small open models at the cost of BERTScore. In domain it is more accurate on the shortcut-controlled axis than every other reference-free metric we could run, including a local LLM judge that costs 82 times more, and its consistency scorer transfers to FRANK, where it matches AlignScore. Its novelty is empirical rather than conceptual: every signal it uses has prior art, and the combination is a calibrated ridge regression. What is new is the choice of signals by their behaviour on the partialled axis, the per-dimension calibration, and the evidence about where such a calibration does and does not transfer.

What LNC is not. LNC is not a replacement for UniEval when a reference exists: UniEval is better on coherence and fluency, uses a reference for relevance, and costs only 1.6 times more. It is not a general relevance metric: its relevance scorer does not transfer to content-coverage annotations. And it is not independent of SummEval: part of its calibration is specific to the

systems it was fitted on, and a ridge over existing reference-based metrics reaches the same in-domain score. We therefore present it as a cheap reference-free metric for consistency, fluency and coherence of news summaries, with relevance as its weak dimension.

Recommendations for metric evaluation. Three practices would have changed our own conclusions had we adopted them earlier. First, report correlations with extractiveness partialled out next to the raw ones, and include a lead-window control and the copy rate itself in every comparison; on SummEval they beat most metrics on the raw axis. Second, evaluate any metric that is calibrated on human ratings on held-out articles and, if possible, held-out systems and corpora; the dataset-order split of SummEval was favourable to LNC, and unseen systems were harder for it. Third, fix the transfer protocol before running it, and report the verdict even when it is negative.

8.1 Limitations

Baselines below their published strength. Our G-Eval judge is a quantised 7B model with three samples, not GPT-4 with twenty, and GPTEval uses GPT-2 small. Both are therefore weaker than in their original papers, and the comparisons with them characterise cheap local configurations, not the methods in general. Our UniEval omits the sentence-level aggregation of the original for consistency, which may explain part of its weakness on that dimension. AlignScore uses the official weights and is the fairest of the strong baselines.

One calibration corpus and one language. LNC is calibrated on 800 English news summaries rated by three experts each. Its features are language-specific in parts (the perturbation rules and the entity heuristic), and its transfer was tested only on English news and XSum.

The partial axis is a single control. Partialling out the bigram copy rate removes one linear, rank-level confound. It does not remove nonlinear dependence on copying or other confounds such as length, and the copy rate is itself correlated with quality: a summary that copies accurately is often a good summary. The partial axis is a diagnostic of which part of a metric’s agreement is not explained by copying, not a definition of quality.

Cost on one machine. Wall-clock cost depends on the hardware, batching and runtime. We measured every metric on the same machine with the models preloaded, which makes the comparison internally fair, but absolute numbers will differ elsewhere. The ratios we emphasise span one to four orders of magnitude, larger than the variation between typical GPU setups.

No downstream validation. We did not test LNC as a reward for training, as a selection criterion between systems in deployment, or for the interpretability of its feature weights by human readers. Its system-level correlations (Appendix A) are high but rest on 16 systems.

8.2 Threats to validity

The SummEval development half was used for every design decision, but the signals that LNC combines were found during a search in which we looked at full-set correlations of earlier metrics, so the held-out half is not entirely untouched by our choices. The random-split and unseen-system analyses, which refit the calibration from scratch, and the frozen transfer test bound the effect of this. The transfer corpora were converted to SummEval’s layout by our own script; we release it, and it rebuilds the files byte for byte from their public sources. All reported numbers are generated by scripts from stored per-sample scores, and a second run of the whole pipeline on the same machine reproduced the scores of LNC and of every deterministic metric exactly.

9 Conclusion

LNC shows that two small open models are enough for a reference-free metric that covers all four dimensions of summary quality: a calibrated combination of language-model likelihood, perturbation contrasts and sentence-level entailment is cheaper than BERTScore and more accurate than a local LLM judge, and it keeps most of its accuracy when the extractive shortcut of the benchmark is controlled for. A pre-registered test shows that its consistency judgement transfers to new systems and annotation schemes while its relevance judgement does not, and controls show where its in-domain accuracy comes from. We hope that the evaluation protocol, with shortcut-controlled correlations, same-machine costs, frozen transfer and calibrated controls, proves as useful as the metric.

Data and code availability

All code, the four corpora in a common format, the per-sample scores of every metric, the frozen LNC calibration and the scripts that render every table in this paper are available at <https://github.com/mikolajsemeniuk/llmbench>. The command `make clean data reproduce` rebuilds the corpora from their public sources and regenerates every result. SummEval, FRANK and RoSE are distributed by their authors under their own licences.

CRedit authorship contribution statement

Mikolaj Semeniuk: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft. **Agnieszka Jastrzebska:** Conceptualization, Methodology, Validation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with writing analysis code and with drafting and editing the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

A Additional tables

This appendix gives the remaining tables rendered by the analysis scripts. Table 16 lists the raw summary-level Spearman correlation of every metric with every dimension on the held-out SummEval articles, with 95% bootstrap intervals over articles. Table 17 gives the system-level

Table 16: Summary-level correlations with human judgment on SummEval (held-out last 50 articles). ρ = Spearman. Brackets give the 95% bootstrap confidence interval over articles.

Metric	Coh	Con	Flu	Rel
BLEU	.079 [-.007, .162]	.116 [.036, .183]	.053 [-.022, .133]	.220 [.107, .320]
ROUGE-L	.140 [.049, .232]	.130 [.050, .206]	.113 [.033, .193]	.176 [.059, .281]
ChrF	.130 [.042, .218]	.136 [.055, .209]	.123 [.029, .208]	.268 [.140, .376]
METEOR	.177 [.100, .254]	.078 [-.002, .152]	.082 [-.003, .175]	.161 [.065, .258]
SMART-String	-.034 [-.114, .047]	.068 [-.008, .140]	.101 [.028, .179]	.091 [-.019, .194]
EmbedScorer	.221 [.150, .290]	.160 [.083, .237]	.157 [.069, .245]	.350 [.251, .444]
BERTScore	.421 [.347, .485]	.107 [.030, .180]	.130 [.060, .207]	.418 [.335, .499]
MoverScore	.187 [.103, .269]	-.019 [-.111, .070]	.015 [-.065, .105]	.173 [.080, .259]
SMART-Model	.156 [.071, .239]	.151 [.073, .229]	.133 [.045, .234]	.250 [.133, .356]
BARTScore	.093 [.004, .190]	.046 [-.021, .123]	.126 [.044, .206]	.197 [.072, .320]
GPTScore	.413 [.337, .488]	.231 [.163, .296]	.179 [.098, .271]	.339 [.258, .417]
UniEval	.580 [.532, .625]	.404 [.350, .459]	.459 [.403, .512]	.448 [.374, .514]
G-Eval	.360 [.289, .429]	.500 [.440, .557]	.122 [.058, .183]	.328 [.242, .402]
AlignScore	.193 [.114, .272]	.427 [.364, .486]	.372 [.308, .433]	.134 [.067, .211]
NLIG	.183 [.113, .251]	.468 [.407, .526]	.280 [.214, .346]	.164 [.091, .240]
Lead-3	.269 [.192, .345]	.330 [.280, .384]	.234 [.163, .301]	.245 [.149, .330]
Lead-5	.287 [.210, .365]	.371 [.327, .414]	.243 [.171, .315]	.268 [.194, .336]
Lead-3 (whole)	.255 [.179, .334]	.294 [.235, .353]	.220 [.149, .291]	.276 [.183, .367]
LGS	.299 [.227, .369]	.363 [.313, .411]	.252 [.183, .319]	.281 [.195, .352]
CCM	.251 [.182, .324]	.324 [.257, .390]	.233 [.171, .299]	.185 [.114, .252]
PSC	.269 [.189, .340]	.260 [.196, .331]	.205 [.137, .269]	.209 [.126, .279]
SPL	.122 [.044, .199]	.026 [-.045, .098]	.143 [.078, .206]	.076 [.002, .162]
LNC	.406 [.342, .478]	.514 [.466, .560]	.431 [.384, .479]	.379 [.302, .440]

correlations on the same articles, computed from the mean score of each of the 16 systems; with 16 systems their intervals are wide. Table 18 gives the paired comparisons on FRANK XSum.

Reproduction. Every table in this paper is written by a command of the released code: `make paper` renders all of them from the stored per-sample scores without any model, and `make reproduce` recomputes the scores first (about 16 hours on the machine described in Section 4, half of it for G-Eval).

Table 17: System-level correlations with human judgment on SummEval (held-out last 50 articles). ρ = Spearman. Brackets give the 95% bootstrap confidence interval over systems.

Metric	Coh	Con	Flu	Rel
BLEU	-.071 [-.702, .520]	-.074 [-.596, .468]	.053 [-.543, .596]	.206 [-.484, .679]
ROUGE-L	.174 [-.486, .698]	-.024 [-.618, .499]	.359 [-.244, .763]	.447 [-.208, .833]
ChrF	.259 [-.283, .719]	.737 [.335, .912]	.449 [-.120, .799]	.521 [.096, .792]
METEOR	-.047 [-.629, .580]	-.383 [-.805, .227]	.056 [-.454, .617]	.115 [-.482, .687]
SMART-String	-.129 [-.827, .488]	-.265 [-.749, .294]	.082 [-.513, .617]	.159 [-.528, .680]
EmbedScorer	.368 [-.139, .717]	.864 [.565, .956]	.572 [.066, .824]	.647 [.292, .798]
BERTScore	.615 [.051, .879]	-.086 [-.629, .489]	.372 [-.203, .814]	.362 [-.321, .830]
MoverScore	.121 [-.506, .686]	-.649 [-.873, -.201]	-.287 [-.757, .309]	-.076 [-.681, .560]
SMART-Model	.138 [-.373, .639]	.770 [.458, .917]	.574 [.068, .844]	.444 [-.093, .808]
BARTScore	.294 [-.277, .734]	.681 [.277, .898]	.375 [-.209, .763]	.526 [.091, .821]
GPTScore	.624 [.036, .943]	.457 [-.107, .907]	.372 [-.291, .833]	.350 [-.291, .869]
UniEval	.815 [.454, .979]	.440 [-.195, .850]	.705 [.258, .939]	.803 [.434, .935]
G-Eval	.565 [.010, .895]	.970 [.836, .998]	.348 [-.195, .830]	.751 [.295, .954]
AlignScore	.379 [-.191, .792]	.558 [-.058, .896]	.655 [.070, .882]	.618 [.076, .892]
NLIG	.226 [-.354, .698]	.873 [.562, .986]	.505 [-.084, .832]	.456 [-.207, .902]
Lead-3	.159 [-.405, .743]	.785 [.406, .956]	.422 [-.252, .847]	.341 [-.366, .925]
Lead-5	.124 [-.450, .750]	.767 [.375, .952]	.414 [-.254, .857]	.306 [-.410, .919]
Lead-3 (whole)	.185 [-.355, .757]	.764 [.355, .946]	.452 [-.159, .874]	.344 [-.357, .943]
LGS	.079 [-.488, .707]	.764 [.368, .955]	.369 [-.309, .812]	.282 [-.424, .934]
CCM	.271 [-.364, .805]	.779 [.415, .928]	.419 [-.252, .839]	.447 [-.233, .910]
PSC	.385 [-.211, .885]	.758 [.407, .929]	.528 [-.084, .902]	.544 [-.036, .938]
SPL	.676 [.312, .845]	.180 [-.382, .696]	.293 [-.279, .797]	.415 [-.139, .848]
LNC	.532 [-.048, .893]	.870 [.545, .964]	.550 [-.090, .905]	.641 [.152, .916]

Table 18: Paired bootstrap comparison of LNC against baselines on FRANK (XSum), copy-partialled summary-level Spearman correlations. Each cell reports $\Delta\rho$ (ours minus baseline) with 95% CI in brackets, the raw two-sided p -value, and q , the Benjamini-Hochberg adjusted p -value over all 7 (baseline, dimension) cells of this table. Bold marks improvements that survive the correction ($q < 0.05$ and CI excludes 0).

Baseline	Con
AlignScore	-.013 [-.061, +.034] $p=0.605$ $q=0.605$
NLIG	+.025 [-.019, +.070] $p=0.267$ $q=0.468$
LGS	+.174 [+.090, +.255] $p < .001$ $q < .001$
Lead-3	+.226 [+.152, +.298] $p < .001$ $q < .001$
Lead-5	+.198 [+.121, +.272] $p < .001$ $q < .001$
UniEval	+.017 [-.042, +.071] $p=0.550$ $q=0.605$
G-Eval	+.016 [-.047, +.076] $p=0.594$ $q=0.605$

References

- [1] Reinald Kim Amplayo, Peter J. Liu, Yao Zhao, and Shashi Narayan. SMART: Sentences as basic units for text evaluation. In *Proceedings of the Eleventh International Conference on Learning Representations (ICLR)*, 2023. URL <https://openreview.net/pdf?id=0Ie3kpw140D>.
- [2] Satanjeev Banerjee and Alon Lavie. METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*, pages 65–72, Ann Arbor, Michigan, 2005. Association for Computational Linguistics. URL <https://aclanthology.org/W05-0909/>.
- [3] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- [4] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.
- [5] Daniel Deutsch, Rotem Dror, and Dan Roth. A statistical analysis of summarization evaluation metrics using resampling methods. *Transactions of the Association for Computational Linguistics*, 9:1132–1146, 2021. doi: 10.1162/tac1_a_00417.
- [6] Esin Durmus, Faisal Ladhak, and Tatsunori Hashimoto. Spurious correlations in reference-free evaluation of text generation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1443–1454, 2022. doi: 10.18653/v1/2022.acl-long.102.
- [7] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*, volume 57 of *Monographs on Statistics and Applied Probability*. Chapman & Hall/CRC, New York, 1993. ISBN 9780412042317.
- [8] Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. SummEval: Re-evaluating summarization evaluation. *Transactions of the Association for Computational Linguistics*, 9:391–409, 2021. doi: 10.1162/tac1_a_00373. URL <https://aclanthology.org/2021.tac1-1.24/>.
- [9] Alexander R. Fabbri, Chien-Sheng Wu, Wenhao Liu, and Caiming Xiong. QAFactEval: Improved QA-based factual consistency evaluation for summarization. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2587–2601, 2022. doi: 10.18653/v1/2022.naacl-main.187.
- [10] Milton Friedman. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the American Statistical Association*, 32(200):675–701, 1937. doi: 10.1080/01621459.1937.10503522.
- [11] Jinlan Fu, See-Kiong Ng, Zhengbao Jiang, and Pengfei Liu. GPTScore: Evaluate as you desire. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6556–6576, Mexico City, Mexico, 2024. doi: 10.18653/v1/2024.naacl-long.365. URL <https://aclanthology.org/2024.naacl-long.365/>.
- [12] Yang Gao, Wei Zhao, and Steffen Eger. SUPERT: Towards new frontiers in unsupervised evaluation metrics for multi-document summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1347–1354, 2020. doi: 10.18653/v1/2020.acl-main.124.
- [13] Sebastian Gehrmann, Elizabeth Clark, and Thibault Sellam. Repairing the cracked foundation: A survey of obstacles in evaluation practices for generated text. *Journal of Artificial Intelligence Research*, 77:103–166, 2023. doi: 10.1613/jair.1.13715.
- [14] Max Grusky, Mor Naaman, and Yoav Artzi. Newsroom: A dataset of 1.3 million summaries with diverse extractive strategies. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 708–719, 2018. doi: 10.18653/v1/N18-1065. URL <https://aclanthology.org/N18-1065/>.
- [15] Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Saizhuo Wang, Kun Zhang, Yuanzhuo Wang, Wen Gao, Lionel Ni, and Jian Guo. A survey on LLM-as-a-judge. *The Innovation*, 7(6), 2026. URL <https://www.the-innovation.org/article/id/697872b44f6e8d88d813340c>.
- [16] Pengcheng He, Jianfeng Gao, and Weizhu Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2023.
- [17] Karl Moritz Hermann, Tomáš Kočiský, Edward Grefenstette, Lasse Espeholt, Will Kay, Mustafa Suleyman, and Phil Blunsom. Teaching machines to read and comprehend. In *Advances in Neural Information Processing Systems (NIPS)*, volume 28, pages 1693–1701, 2015. URL <https://arxiv.org/abs/1506.03340>.
- [18] Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970. doi: 10.1080/00401706.1970.10488634.
- [19] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2): 65–70, 1979.

- [20] Qi Jia, Siyu Ren, Yizhu Liu, and Kenny Q. Zhu. Zero-shot faithfulness evaluation for text summarization with foundation language model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.
- [21] Chris Kedzie, Kathleen McKeown, and Hal Daumé III. Content selection in deep learning models of summarization. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1818–1828, 2018. doi: 10.18653/v1/D18-1208. URL <https://aclanthology.org/D18-1208/>.
- [22] Wojciech Kryściński, Bryan McCann, Caiming Xiong, and Richard Socher. Evaluating the factual consistency of abstractive text summarization. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9332–9346, 2020. doi: 10.18653/v1/2020.emnlp-main.750. URL <https://aclanthology.org/2020.emnlp-main.750/>.
- [23] Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. *Transactions of the Association for Computational Linguistics*, 10:163–177, 2022. doi: 10.1162/tac1_a_00453. URL <https://aclanthology.org/2022.tac1-1.10/>.
- [24] Faisal Ladhak, Esin Durmus, He He, Claire Cardie, and Kathleen McKeown. Faithful or extractive? on mitigating the faithfulness-abstractiveness trade-off in abstractive summarization. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1410–1421, 2022. doi: 10.18653/v1/2022.acl-long.100.
- [25] Moritz Laurer, Wouter van Atteveldt, Andreu Casas, and Kasper Welbers. Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and BERT-NLI. *Political Analysis*, 32(1):84–100, 2024. doi: 10.1017/pan.2023.20.
- [26] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 7871–7880, 2020. doi: 10.18653/v1/2020.acl-main.703. URL <https://aclanthology.org/2020.acl-main.703/>.
- [27] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, 2004. Association for Computational Linguistics. URL <https://aclanthology.org/W04-1013/>.
- [28] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2511–2522, 2023. doi: 10.18653/v1/2023.emnlp-main.153. URL <https://aclanthology.org/2023.emnlp-main.153/>.
- [29] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. arXiv preprint arXiv:1907.11692, 2019.
- [30] Yixin Liu, Alexander R. Fabbri, Pengfei Liu, Yilun Zhao, Linyong Nan, Ruilin Han, Simeng Han, Shafiq Joty, Chien-Sheng Wu, Caiming Xiong, and Dragomir Radev. Revisiting the gold standard: Grounding summarization evaluation with robust human evaluation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4140–4170, 2023. doi: 10.18653/v1/2023.acl-long.228.
- [31] Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919, 2020. doi: 10.18653/v1/2020.acl-main.173.
- [32] Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D. Manning, and Chelsea Finn. DetectGPT: Zero-shot machine-generated text detection using probability curvature. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 24950–24962, 2023.
- [33] Shashi Narayan, Shay B. Cohen, and Mirella Lapata. Don’t give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 1797–1807, 2018. doi: 10.18653/v1/D18-1206.
- [34] Zach Nussbaum, John X. Morris, Brandon Duderstadt, and Andriy Mulyar. Nomic embed: Training a reproducible long context text embedder. *Transactions on Machine Learning Research*, 2024. URL <https://arxiv.org/abs/2402.01613>.
- [35] Ollama. Ollama: Get up and running with large language models, locally. <https://ollama.com>, 2024. Software project page, accessed 2026-05-31.
- [36] Artidoro Pagnoni, Vidhisha Balachandran, and Yulia Tsvetkov. Understanding factuality in abstractive summarization with FRANK: A benchmark for factuality metrics. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4812–4829, 2021. doi: 10.18653/v1/2021.naacl-main.383.

- [37] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 311–318, 2002. doi: 10.3115/1073083.1073135. URL <https://aclanthology.org/P02-1040/>.
- [38] Maja Popović. chrF: character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation (WMT)*, pages 392–395, 2015. doi: 10.18653/v1/W15-3049. URL <https://aclanthology.org/W15-3049/>.
- [39] Qwen Team. Qwen2.5 technical report. arXiv preprint arXiv:2412.15115, 2024.
- [40] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. OpenAI technical report, 2019.
- [41] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL <https://jmlr.org/papers/v21/20-074.html>.
- [42] Ricardo Rei, Craig Stewart, Ana C. Farinha, and Alon Lavie. COMET: A neural framework for MT evaluation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2685–2702, 2020. doi: 10.18653/v1/2020.emnlp-main.213. URL <https://aclanthology.org/2020.emnlp-main.213/>.
- [43] Thomas Scialom, Sylvain Lamprier, Benjamin Piwowarski, and Jacopo Staiano. Answers unite! unsupervised metrics for reinforced summarization models. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3246–3256, 2019. doi: 10.18653/v1/D19-1320.
- [44] Thibault Sellam, Dipanjan Das, and Ankur P. Parikh. BLEURT: Learning robust metrics for text generation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 7881–7892, 2020. doi: 10.18653/v1/2020.acl-main.704. URL <https://aclanthology.org/2020.acl-main.704/>.
- [45] Liyan Tang, Philippe Laban, and Greg Durrett. MiniCheck: Efficient fact-checking of LLMs on grounding documents. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.
- [46] Brian Thompson and Matt Post. Automatic machine translation evaluation in many languages via zero-shot paraphrasing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 90–121, 2020. doi: 10.18653/v1/2020.emnlp-main.8.
- [47] Oleg Vasilyev and John Bohannon. ESTIME: Estimation of summary-to-text inconsistency by mismatched embeddings. In *Proceedings of the 2nd Workshop on Evaluation and Comparison of NLP Systems*, pages 94–103, 2021. doi: 10.18653/v1/2021.eval4nlp-1.10.
- [48] Oleg Vasilyev, Vedant Dharnidharka, and John Bohannon. Fill in the BLANC: Human-free quality estimation of document summaries. In *Proceedings of the First Workshop on Evaluation and Comparison of NLP Systems*, pages 11–20, 2020. doi: 10.18653/v1/2020.eval4nlp-1.2.
- [49] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945. doi: 10.2307/3001968.
- [50] Genta Indra Winata, David Anugraha, Lucky Susanto, Garry Kuwanto, and Derry Tanti Wijaya. MetaMetrics: Calibrating metrics for generation tasks using human preferences. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
- [51] Zhuohan Xie, Trevor Cohn, and Jey Han Lau. DeltaScore: Fine-grained story evaluation with perturbations. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.
- [52] Weizhe Yuan, Graham Neubig, and Pengfei Liu. BARTScore: Evaluating generated text as text generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 27263–27277, 2021. URL <https://arxiv.org/abs/2106.11520>.
- [53] Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. AlignScore: Evaluating factual consistency with a unified alignment function. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11328–11348, 2023. doi: 10.18653/v1/2023.acl-long.634.
- [54] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR)*, 2020. URL <https://openreview.net/forum?id=SkeHuCVFDr>.
- [55] Wei Zhao, Maxime Peyrard, Fei Liu, Yang Gao, Christian M. Meyer, and Steffen Eger. MoverScore: Text generation evaluating with contextualized embeddings and Earth Mover Distance. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 563–578, 2019. doi: 10.18653/v1/D19-1053. URL <https://aclanthology.org/D19-1053/>.
- [56] Ming Zhong, Yang Liu, Da Yin, Yuning Mao, Yizhu Jiao, Pengfei Liu, Chenguang Zhu, Heng Ji, and Jiawei Han. Towards a unified multi-dimensional evaluator for text generation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2023–2038, 2022. doi: 10.18653/v1/2022.emnlp-main.131. URL <https://aclanthology.org/2022.emnlp-main.131/>.